JOURNAL
OF HEPATOLOGY

Hepatocellular carcinoma in patients with autoimmune hepatitis

To the Editor:

We read the interesting article by Colapietro *et al.* published in *Journal of Hepatology* reporting – in a multicentre, international study – that the incidence of hepatocellular carcinoma (HCC) in patients with autoimmune hepatitis (AIH) is low (0.09% per year in the first 10 years), and remains low (0.46% per year in the first 10 years) even considering patients with cirrhosis alone.[1] As emphasised in the editorial by Grønbæk *et al.*, these findings confirm that the occurrence of HCC in patients with AIH is a rare event, and corroborate previous results from single-centre studies.[2–4] The low incidence of HCC puts into question whether, from a cost-effective standpoint, patients with AIH, even with cirrhosis, should undergo surveillance, considering that the figure reported is well below the threshold for a cost-effective surveillance (*i.e.*, 1.5% per year).[1,5,6]

Due to these updated findings, Colapietro *et al.* suggest that surveillance for HCC should be personalised in patients with AIH focussing on those at highest risk, such as those with additional independent risk factors, represented by cirrhosis, obesity, and AIH/primary sclerosing cholangitis overlap syndrome.[1] However, the role of overlap syndrome needs to be interpreted with caution as the diagnosis of HCC in these patients was confirmed by histology only in half of the cases, with potential misdiagnosis of intrahepatic cholangiocarcinoma.

We feel that these new data are important, as in the near future the population of patients with HCC will be enriched for virus-negative cases that will not all be secondary to metabolic disorders.[7,8] Therefore, adequate characterisation of patients with rare conditions, such as AIH, who develop HCC may provide meaningful information regarding prevention and management.[2,9]

In this regard, we deemed it of interest to assess the characteristics of patients with AIH and HCC included in the Italian Liver Cancer (ITA.LI.CA) database. Among 8,038 patients with HCC diagnosed between 2009 and 2022, we identified 18 patients (0.2%) with AIH and HCC whose main characteristics are shown in Table 1. Male patients represented half of cases, all patients were ≥50 years old, and 13 patients (72.2%) were ≥65 years old. Seven patients (38.9%) were either overweight or obese, cirrhosis was present in 16 patients (90.0%) and in the two patients without cirrhosis this finding was histologically confirmed as both underwent surgery. Two patients (11.1%) had AIH/primary biliary cholangitis overlap syndrome, and three patients (16.7%) reported alcohol abuse. The majority of patients had well-compensated liver disease, and 55.6% of HCC diagnoses were made during surveillance. Overall, 72.2% of patients had tumours within the Milan criteria, none had macrovascular invasion, while three patients (16.7%) presented extrahepatic spread at diagnosis. Due to these favourable characteristics, the most frequent initial treatments for HCC were potentially curative (61.1%), with six patients (33.3%) undergoing ablation, four patients (22.2%) liver resection, and one patient (5.6%) liver transplantation. Only one patient was ineligible for any treatment and was managed by best supportive care. The association of preserved liver function, favourable oncologic stage and dominant curative treatment led to a median overall survival of 41.7 months (95% CI 5.7–77.7).

Table 1. Characteristics of patients with autoimmune hepatitis and hepatocellular carcinoma.

Characteristics	Data
Gender, male	9 (50.0)
Age, years	67.2 (64.6–71.5)
Body mass index, kg/m^2	26.2 (24.7–33.2)
Platelet count, x10^9/L	77 (59–89)
Child-Turcotte-Pugh score	
A	10 (55.6)
B	7 (38.9)
C	1 (5.6)
Number of nodules	
1	14 (77.8)
2-3	3 (16.7)
>3	1 (5.6)

Data are shown as median and interquartile range, or as absolute number and percentage.

How can these results be put into perspective with those of Colapietro *et al.*?[1] First, despite their study being unable to identify an increased risk of HCC associated with male sex, we observed that half of our patients with AIH and HCC were males, a finding previously reported in studies carried out in the US and UK, thus highlighting the potentially higher risk in a gender where AIH is usually underrepresented.[3,4] Second, our data emphasise once more the relevance of cirrhosis as an oncologic risk, since this condition was found in 90% of our patients, a finding in line with both older (100%, 1971-2007) and more recent (90%, 2000-2014) series across the globe.[3,4] Third, in approximately one-third of our patients there were concomitant risk factors such as primary biliary cholangitis and alcohol abuse, and about 40% of patients had concomitant metabolic risk factors.[9,10]

Our findings, besides providing a well-characterised picture of patients with AIH and HCC – *i.e.* elderly males with cirrhosis and concomitant multiple aetiologic risk factors for HCC – emphasise the potential utility of HCC surveillance for these individuals, who are usually on-treatment for AIH and have well-compensated liver disease. Indeed, a favourable clinical and oncologic stage at diagnosis allows us to use curative treatments, eventually leading to a median survival of approximately 4 years.

To conclude, our data confirm that HCC in AIH is a rare event, prevalently occurring in older, male patients with cirrhosis, who frequently have additional risk factors for HCC. We also provided evidence that the prognosis of patients with AIH and HCC is good if they can access curative treatments due to limited tumour burden. Therefore, future studies should be aimed at providing suitable calculators of HCC risk to select, among patients with AIH, those in whom the incidence of HCC crosses the threshold for cost-effective surveillance.

Andrea Pasta[1]
Giulia Pieri[1,2]
Maria Corina Plaz Torres[1,2]
Franco Trevisani[3,4]
Edoardo G. Giannini[1,2,*]
for the Italian Liver Cancer (ITA.LI.CA) Group
[1]*Gastroenterology Unit, Department of Internal Medicine, University of Genoa, Genoa, Italy*
[2]*Gastroenterology Unit, IRCCS Ospedale Policlinico San Martino, Genoa, Italy*
[3]*Unit of Semeiotics, Liver and Alcohol-related Diseases, University of Bologna, Bologna, Italy*
[4]*IRCCS Azienda Ospedaliero-Universitaria di Bologna, Bologna, Italy*
[*]Corresponding author. Address: Gastroenterology Unit, Department of Internal Medicine, University of Genoa, Viale Benedetto XV, no. 6, 16132, Genoa, Italy.
E-mail address: egiannini@unige.it (E.G. Giannini)

Received 26 February 2024; Received in revised form 10 March 2024; Accepted 19 March 2024; Available online 26 March 2024
https://doi.org/10.1016/j.jhep.2024.03.025

Financial support

The authors received no financial support to produce this manuscript.

Conflict of interest

The authors of this study declare that they do not have any conflict of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Giannini EG: Conceptualization; Writing - original draft, review & editing. Pasta A, Pieri A, Plaz Torres MC: Writing - review & editing. Trevisani F: Supervision; Writing - review & editing.

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhep.2024.03.025.

References

Author names in bold designate shared co-first authorship

[1] Colapietro F, Maisonneuve P, Lytvyak E, et al. Incidence and predictors of hepatocellular carcinoma in patients with autoimmune hepatitis. J Hepatol 2024;80:53–61.
[2] Grønbæk L, Grønbæk H. Hepatocellular carcinoma in autoimmune hepatitis remains rare. J Hepatol 2024;80:8–9.
[3] Dakhoul L, Jones KR, Gawrieh S, et al. Older age and disease duration are highly associated with hepatocellular carcinoma in patients with autoimmune hepatitis. Dig Dis Sci 2019;64:1705–1710.
[4] Yeoman AD, Al-Chalabi T, Karani JB, et al. Evaluation of risk factors in the development of hepatocellular carcinoma in autoimmune hepatitis: implications for follow-up and screening. Hepatology 2008;48:863–870.
[5] Tansel A, Katz LH, El-Serag HB, et al. Incidence and determinants of hepatocellular carcinoma in autoimmune hepatitis: a systematic review and meta-analysis. Clin Gastroenterol Hepatol 2017;15:1207–1217.e4.
[6] Giannini EG, Cucchetti A, Erroi V, et al. Surveillance for early diagnosis of hepatocellular carcinoma: how best to do it? World J Gastroenterol 2013;19:8808–8821.
[7] Garuti F, Neri A, Avanzato F, et al. The changing scenario of hepatocellular carcinoma in Italy: an update. Liver Int 2021;41:585–597.
[8] **Vitale A**, **Svegliati-Baroni G**, Ortolani A, et al. Epidemiological trends and trajectories of MAFLD-associated hepatocellular carcinoma 2002-2033: the ITA.LI.CA database. Gut 2023;72:141–152.
[9] Giannini EG, Pieri G, Labanca S, et al. Characteristics and survival of patients with primary biliary cholangitis and hepatocellular carcinoma. Dig Liver Dis 2022;54:1215–1221.
[10] Reggidori N, Bucci L, Santi V, et al. Landscape of alcohol-related hepatocellular carcinoma in the last 15 years highlights the need to expand surveillance programs. JHEP Rep 2023;5:100784.